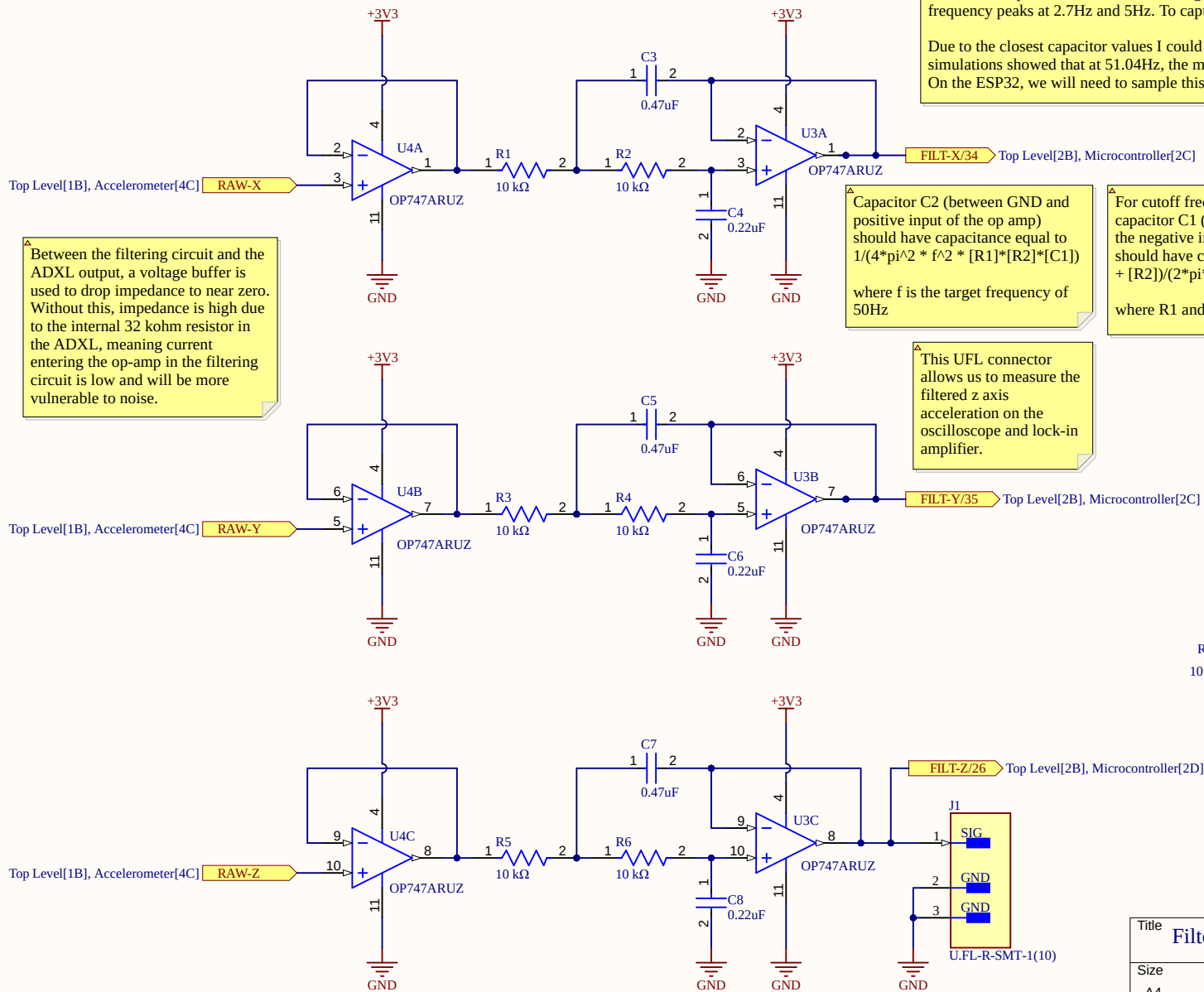




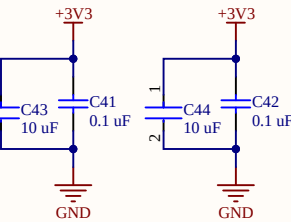
Between the filtering circuit and the ADXL output, a voltage buffer is used to drop impedance to near zero. Without this, impedance is high due to the internal 32 kohm resistor in the ADXL, meaning current entering the op-amp in the filtering circuit is low and will be more vulnerable to noise.

This is a low-pass Butterworth filter using Sallen-Key topology. A running test with phyphox showed frequency peaks at 2.7Hz and 5Hz. To capture extra harmonics, we made the cutoff $5 \times 10 = 50\text{Hz}$.
Due to the closest capacitor values I could find, its theoretical cutoff is around 49.49Hz. LTspice simulations showed that at 51.04Hz, the magnitude is -3dB. At 20Hz, the phase angle is approx. -33.46 deg. On the ESP32, we will need to sample this filtered signal at a minimum sampling frequency of 100Hz.

Capacitor C2 (between GND and positive input of the op amp) should have capacitance equal to $1/(4 \times \pi^2 \times f^2 \times [R1] \times [R2] \times [C1])$ where f is the target frequency of 50Hz

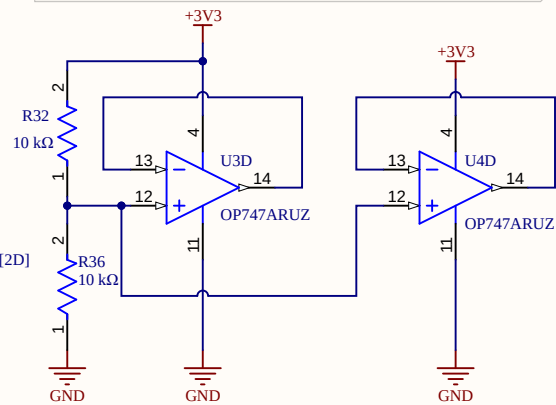
For cutoff frequency of 50Hz, capacitor C1 (between RAW-X and the negative input of the op amp) should have capacitance equal to $([R1] + [R2]) / (2 \times \pi \times f \times [R1] \times [R2] \times \text{sqrt}(2))$ where R1 and R2 = 10 kΩ

This UFL connector allows us to measure the filtered z axis acceleration on the oscilloscope and lock-in amplifier.



C41 and C43 are the decoupling capacitors for U4, C42 and C44 are the decoupling capacitors for U3.

Each OP747 chip features four op amps. Since we only need 6 for x, y and z, there are 2 unused ones. Output is looped into V-, and V+ receives +1.65V from a voltage divider to properly terminate the op amp as suggest in TI's SBOA204A document



Title Filtering Circuit		
Size A4	Number LUM-2026-SCH-003	Revision v1.0
Date: 5/22/2026	Sheet 3 of 9	
File: C:\Users\...\Filter.SchDoc	Drawn By: Leo Chinchilla	